Robustness of Implicit Q-Learning Under Controlled Distribution Shift

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Core Challenge

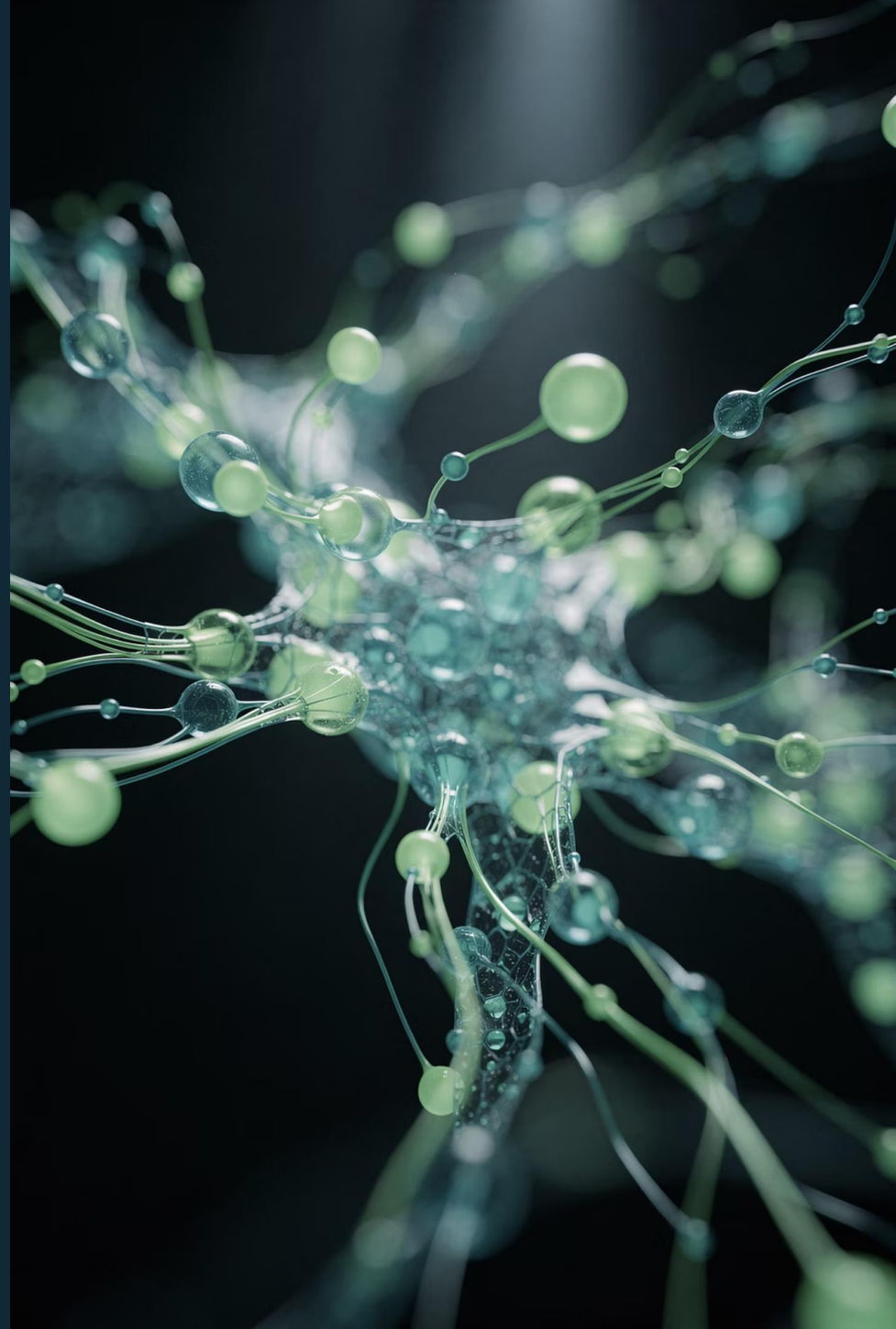
Offline RL must generalize under shift.

Algorithm Studied

IQL uses expectile and advantage-weighted updates.

Key Extension

TripleCritic and τ ablation study.



Background: Offline RL & Distribution Shift

Why Offline RL?

It is useful in safety-critical settings where interaction is costly or impossible. Models train on fixed datasets but must still generalize.

Research Question

How robust is IQL under controlled distribution shift?

Three Competing Strategies

→ CQL

Uses pessimism to reduce out-of-distribution Q-values.

→ IQL

Stays within the dataset support for updates.

→ TD3+BC

Uses behavior cloning to keep the policy close.

Methodology: IQL Reproduction & Reproduction & Shift Wrappers Wrappers

We train IQL once on standard D4RL medium datasets for 300k steps, then apply then apply wrappers only at test time.

Gravity Wrapper

Scales gravity from 0.5× to 2.0×.

Observation Noise

Adds Gaussian noise at four levels.

Friction Wrapper

Scales friction from 0.5× to 2.0×.

Reward Perturbation

Adds reward noise as the control.

Experimental Results: 2Q vs. 3Q Robustness

Baseline Raw Returns (τ=0.7)

Environment	2Q (Baseline)	3Q (Ensemble)
hopper-medium-v2	1571 ± 136	1469 ± 38
halfcheetah-medium-v2	5543 ± 31	5501 ± 35
walker2d-medium-v2	3360 ± 152	3423 ± 104

Hopper AUDC

Shift Type	2Q	3Q	Winner
Gravity	0.616 ± 0.027	0.529 ± 0.069	**3Q**
Obs Noise	0.134 ± 0.020	0.126 ± 0.017	**3Q**
Friction	0.713 ± 0.026	0.692 ± 0.011	**3Q**
Reward Perturb	0.002 ± 0.001	0.001 ± 0.000	**3Q**

AUDC Results: All Environments

Gravity AUDC (lower = more robust) across Hopper, HalfCheetah, Walker2d — mean ± std, 4 seeds

Hopper

Gravity

2Q: 0.616 ± 0.027

3Q: 0.529 ± 0.069 ✓

Obs. Noise

2Q: 0.134 3Q: 0.126 ✓

Friction

2Q: 0.713 3Q: 0.692 ✓

HalfCheetah

Gravity

2Q: 0.255 ± 0.021

3Q: 0.258 ± 0.004 ~Tie

Obs. Noise

2Q: 0.142 3Q: 0.135 ✓

Friction

2Q: 0.016 ~Tie 3Q: 0.018

Walker2d

Gravity

2Q: 0.716 ± 0.024 ✓

3Q: 0.739 ± 0.038

Obs. Noise

2Q: 0.122 3Q: 0.107 ✓

Friction

2Q: 0.109 ✓ 3Q: 0.131

Ablation Study: Expectile τ & Gravity Case Study

Hopper Gravity Shift

Gravity	2Q Return	2Q Drop%
0.5×	430	72.6%
1.0×	1571	0.0%
1.5×	830	47.2%
2.0×	440	72.0%

Hopper AUDC by Expectile τ (2Q)

τ	Baseline	Gravity AUDC
0.5	1529	0.538
0.7	1571	0.616
0.8	1899	0.730
0.9	1928	0.687

Key Findings & Future Directions

Key Findings

Finding	Evidence
3Q helps Hopper only	Gravity AUDC: 0.529 (3Q) vs. 0.616 (2Q)
Lower τ = more robust	τ=0.5 gravity AUDC 0.538 vs. τ=0.8 at 0.730
Reward shift negligible	AUDC < 0.002 across all envs

Future Directions

Direction	Rationale
Shift-Aware τ Selection	τ trade-off varies by deployment shift type
Adaptive Ensemble Sizing	3Q inconsistent across Walker2d & HalfCheetah

Optimal (critic count, τ) pair is environment-dependent — no single config dominates across all tasks.

Code: github.com/shloakk/iql-robustness-analysis